CHEM 101 Report for Laboratory Exercise #2

Instructions:

a) Download the .docx file (do not download the pdf file from Brightspace). Using Microsoft Word or equivalent software, insert responses in all yellow highlighted areas.

b) It is recommended the report be completed without changing font size, column width, row width, margins, and highlights. Keep the yellow highlighted areas.

c) Additional instructions are provided throughout the report in the gray highlighted areas. Please read and follow these instructions.

d) All answers must be the student's own work without assistance from others.

e) The completed report must be uploaded to the CHEM 101 Brightspace site as a .pdf file by the due date posted on Brightspace. The report is due to 48 hours of the end of the laboratory period. Late reports will NOT be accepted.

f) ***Only typed reports completed using the template will be graded****. Handwritten reports (or excerpts) will NOT be accepted or graded.*

	Grade Scheme	**max**
Pre-lab quiz	ALL questions answers are correct	2
Laboratory Notebook:	ALL data, procedures and observations accurately recorded	2

Report	Abstract is succinct, and contains objective of the experiment, methods used and some conclusions.	1
	Data, Results followed the outlined in the in-lab notes. Graph, Tables, Calculations, Reported significant figures,	2
	Algebraic equations are correct.	1
	Discussion questions are ALL completed and answered accurately.	3
	Conclusions (a) summarize the most important findings of the exercise; (b) techniques used; (c) interpretation of their significance; (d) addresses the objective.	1
	References: section is complete and correctly formatted, and in-text citation is correct.	1
	Participation: Understand the procedure and engage with the content.	0.5
	Performance: Student followed the safe practices guidelines without exception. Drawers checked. Submission confirmation shown.	0.5
	Maximum Report Grade	10

© 2025, Department of Chemistry, Faculty of Science

CHEM101 Report for Laboratory Exercise #2

Physical and Chemical Separatory Methods for Heterogeneous and Homogeneous Mixtures

Additional Instruction: Add in-text citation in adequate locations throughout your report. Citations are numbered in order of appearance in text, and references are added at the end of the report.

Name: Evan O'Toole **Lab Section:** B07 **Quad:** 3 **Date:** September 24, 2025

Abstract

The objective of this experiment was to separate both a heterogeneous mixture of NaCl, SiO_2, and $CaCO_3$, and a homogeneous aqueous dye mixture. Physical methods such as dissolution, filtration, and decanting were combined with chemical reactions involving HCl and Na_2CO_3 to isolate the solids. Liquid–liquid extraction with ethyl acetate was used to separate food dyes based on pH-dependent solubility. The heterogeneous mixture was recovered with 72.98% yield, and the dye partitioning results demonstrated the role of pH in controlling solubility and phase partitioning.

Data/Results

Additional Instruction: The observed data inserted in the tables below must be consistent with the observed data written in your laboratory notebook with the correct units.

Table 1. Experimental results of masses obtained after the separation methods. Sample ID #2

	Chemical formula	Mass (g)	% composition in the mixture
Solid A	NaCl	0.2180	11.9%
Solid B	SiO_2	0.8150	44.4%
Solid C	$CaCO_3$	0.8010	43.7%

Table 2. Total mass recovered, original mass (initial) and %recovery of the heterogeneous mixture.

	Results
Original mass (g)	2.513
Total mass recovered (g)	1.834
% recovery	72.98%

Table 3. Colors observed and identification of each layer for the liquid-liquid extraction.

	Lower layer	Upper layer
Initial solution and ethyl acetate	purple	clear
Initial solution and ethyl acetate, after adding HCl.	purple	Tinted purple
After removing the lower layer and adding distilled water.	more strongly tinted purple	Tinted purple

Biphasic system above (ethyl acetate and water) after adding NaOH.	more strongly tinted purple	Tinted purple

Algebraic Equations

Additional Information: See Chapter IV of the CHEM 101 lab manual

1) Mass % composition of the component in the heterogeneous mixture

%Composition (c) = (m_c (g)/$m_{original}$ (g)) x 100%

Where:
c = chemical formula of the recovered component whose composition is being found
m_c (g) = mass of the specified component c
$m_{original}$ (g) = mass of the original mixture
% Composition (c) = % composition of the component c in the heterogenous mixture

2) Mass % of total recovered from the heterogeneous mixture (%yield)

%Recovery= (m_{recov} (g)/ $m_{original}$ (g)) x 100%

Where:
m_{recov}(g) = total mass recovered
$m_{original}$(g)) = total mass of the original

Discussion

Part I – Separation of a Heterogeneous Mixture

1) Describe the methods used to separate solids "A", "B" and "C". (assuming A = NaCl, B = SiO_2, C = $CaCO_3$)

To separate NaCl, water was added to the heterogeneous mixture to dissolve the only water-soluble compound, NaCl[1]. The remaining insoluble solids were removed by gravity filtration, leaving NaCl in the filtrate. The NaCl was then isolated by evaporating the water on a hot plate until dry.

To separate SiO_2, hydrochloric acid (HCl) was added to the remaining solid mixture (SiO_2 and $CaCO_3$). The HCl reacted with $CaCO_3$ to produce CO_2(g), H_2O(l), and $CaCl_2$(aq), leaving SiO_2 as

the only solid[1]. The solid SiO_2 was separated from the supernatant by decanting, and then heated on a hot plate until dry

To separate $CaCO_3$, sodium carbonate (Na_2CO_3) was added to the supernatant containing $CaCl_2$(aq). This reaction produced NaCl(aq) and $CaCO_3$(s). The $CaCO_3$ was collected by vacuum filtration and then dried on a hot plate to isolate the solid[1].

2) Write the balanced equations for the chemical separation method as follows. Use subscripts for indices in chemical formulas and superscripts for charges.

a) solid compound reacted with $HCl_{(aq)}$.

$2HCl_{(aq)} + CaCO_{3(s)} \Rightarrow CO_{2g} + CaCl_{2(aq)} + H_2O_{(l)}$

b) aqueous compound reacted with $Na_2CO_{3(aq)}$.

$CaCl_{2(aq)} + Na_2CO_{3(aq)} \Rightarrow CaCO_{3(s)} + 2NaCl_{(aq)}$

3) Discuss the % recovery obtained and justify your results.

The % recovery obtained was 72.98%, meaning that most, but not all, of the original mixture was recovered in the final product. Losses likely occurred during the two filtration steps, where it is difficult to transfer all of the solid from the filter paper. Additional mass loss may have resulted from small amounts of solid adhering to the glassware during transfers.

Part II – Liquid-Liquid Extraction (Solubility as a function of the solvent)

4) Discuss the importance of controlling the pH in each extraction step and explain how the addition of acid and base influenced the extraction process.

It was important to control the pH in each extraction step, as this altered the solubility of the dyes in water and the organic solvent. Improper pH control could have led to unexpected outcomes. The addition of acid and base influenced solubility by changing the protonation state of the dye molecules: acid protonated certain groups, reducing polarity and making the dyes more soluble in the organic solvent, while base deprotonated these groups, increasing polarity and making the dyes more water-soluble[1].

5) Based on your experimental results, and Figure 2.1 of the lab manual, identify which compound preferentially portioned into the organic layer and which remained in the aqueous layer during the acidic step.

Erythrosine B (Red 3) preferentially partitioned into the organic layer after acidifying the mixture because its carboxyl functional groups were protonated, making it less polar and more soluble in the organic solvent[1]. Alternatively, Brilliant Blue FCF (Blue 1) was less affected by the acid and remained predominantly in the aqueous layer due to its polar groups.

6) Discuss the outcome after the addition of distilled water and base (NaOH) to the separatory funnel.

After adding distilled water and NaOH to the separatory funnel, the base deprotonated the carboxyl functional groups of the remaining Erythrosine B (Red 3) dye. This increased its polarity, making it more water-soluble and causing it to return to the aqueous phase[1].

Conclusions

In this experiment, a heterogeneous mixture of NaCl, SiO_2 and $CaCO_3$ was successfully separated using physical methods such as decanting and filtration. The chemical reactivity of $CaCO_3$ with HCl allowed its separation of the mixture. The %recovery obtained (72.98%) indicates that the process was successful. A homogeneous aqueous solution containing purple food dye was separated using a liquid-liquid extraction procedure. In a separatory funnel after adding a strong acid, Erythosine B (Red 3) was found in the Organic (ethyl acetate) phase, giving it a red color, while Brilliant Blue FCF (Blue 1) was found in the aqueous phase, giving it a blue color. The experiment demonstrated effective separation in homogeneous and heterogeneous mixtures relying in their acid-base properties.

References

Additional Instructions: Note that references must be informed below in the order they are cited in text. Reference #1 is already informed but not yet cited. Add the necessary references below and cite them in text.

1. Lemos M., Reimer M., et al, *Chemistry 101 – Fundamentals of Chemistry from atoms to materials Laboratory Manual*, pp. 33-43. (University of Victoria: Victoria, B.C.). **Fall 2025**.